

Equations with Unknowns on Both Sides

AQA · KS3 Year 8

Name: _____ Class: _____ Date: _____

Solve linear equations where the unknown appears on both sides. Collect the letter terms on one side and the numbers on the other, then solve the simpler equation.

VISUAL EXAMPLE

$$2x + 3 = 11$$

$$-3 \text{ both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



An equation balances: do the same to both sides to keep it level.

COMMON MISTAKES

Forgetting to do the move to both sides — subtract the same letter term from each side.

Moving the larger letter term, which can leave a negative coefficient unnecessarily.

Sign errors when subtracting a term: $5x - 3x = 2x$, not $8x$.

Stopping at $ax = c$ and forgetting the final division.

METHOD

- 1 Decide which side will keep the letters — usually the side with more of them.
- 2 Subtract the smaller letter term from both sides.
- 3 You now have a number on only one side — collect the numbers using inverses.
- 4 Solve the resulting one- or two-step equation.
- 5 Check by substituting into the original equation (both sides should match).

WORKED EXAMPLE

Solve: $5x + 2 = 3x + 10$

Step 1 — Keep letters on the left (it has more x).

Step 2 — Subtract $3x$ from both sides: $2x + 2 = 10$.

Step 3 — Subtract 2 from both sides: $2x = 8$.

Step 4 — Divide by 2: $x = 4$.

Step 5 — Check: $5(4) + 2 = 22$ and $3(4) + 10 = 22$

1 Solve.

(2 marks)

a) $5x = 2x + 9$

b) $7x = 3x + 20$

HOW TO START

Take the smaller letter term away from both sides first.

For $5x = 2x + 9$, subtract $2x$: $3x = 9$, so $x = 3$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Choose the side to keep the letters | 2 Subtract the smaller letter term (both sides) | 3 Collect the numbers using inverses | 4 Solve the simpler equation | 5 Check both sides match

2 Solve.

(2 marks)

a) $6x + 1 = 4x + 9$

b) $8x - 5 = 6x + 7$

HOW TO START

Move the letters one way and the numbers the other.

For $6x + 1 = 4x + 9$, subtract $4x$: $2x + 1 = 9$, then $- 1$: $2x = 8$, $x = 4$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Choose the side to keep the letters | 2 Subtract the smaller letter term (both sides) | 3 Collect the numbers using inverses | 4 Solve the simpler equation | 5 Check both sides match

3 Solve.

(2 marks)

a) $9x - 4 = 4x + 11$

b) $4x + 7 = x + 22$

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Choose the side to keep the letters | 2 Subtract the smaller letter term (both sides) | 3 Collect the numbers using inverses | 4 Solve the simpler equation | 5 Check both sides match

4

(3 marks)

a) $3x + 8 = 7x$

b) $2x - 5 = 5x + 4$

c) $10 - x = 4 + x$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Choose the side to keep the letters | 2 Subtract the smaller letter term (both sides) | 3 Collect the numbers using inverses | 4 Solve the simpler equation | 5 Check both sides match

5 Solve.

(3 marks)

a) $5x + 3 = 2x - 9$

b) $7 - 2x = x + 1$

c) $4x - 1 = 6x - 9$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Choose the side to keep the letters | 2 Subtract the smaller letter term (both sides) | 3 Collect the numbers using inverses | 4 Solve the simpler equation | 5 Check both sides match

6

Sara and Tom have the same number of sweets.

(3 marks)

Sara has $(4x + 2)$ sweets and Tom has $(6x - 8)$ sweets.

- Form an equation. (1 mark)
- Solve it to find x . (1 mark)
- How many sweets does each person have? (1 mark)

ANSWERS

(a) =

(b) =

(c) =

METHOD 1 Choose the side to keep the letters | 2 Subtract the smaller letter term (both sides) | 3 Collect the numbers using inverses | 4 Solve the simpler equation | 5 Check both sides match

7 Two phone plans cost the same for g gigabytes of data. (4 marks)
Plan A: £10 plus £2 per GB. Plan B: £4 plus £4 per GB.

- a) Form an equation in g . (1 mark)
b) Solve it to find the number of GB where the plans cost the same. (2 marks)
c) What is that cost? (1 mark)

ANSWERS

(a) = _____
(b) = _____
(c) = _____

METHOD 1 Choose the side to keep the letters | 2 Subtract the smaller letter term (both sides) | 3 Collect the numbers using inverses | 4 Solve the simpler equation | 5 Check both sides match

8

Two metal rods are the same length.

(5 marks)

Rod A is $(5x - 3)$ cm long and Rod B is $(2x + 9)$ cm long.

- Find x . (2 marks)
- Find the length of each rod. (1 mark)
- A third rod is 5 cm shorter than twice the length of Rod A. Find its length. (2 marks)

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Choose the side to keep the letters | 2 Subtract the smaller letter term (both sides) | 3 Collect the numbers using inverses | 4 Solve the simpler equation | 5 Check both sides match

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

How do I decide which side to keep the letters on?

One thing I will practise before the next lesson:

Teacher key

Q1: a) $3x = 9$ $x = 3$ b) $4x = 20$ $x = 5$

Q2: a) $2x = 8$ $x = 4$ b) $2x = 12$ $x = 6$

Q3: a) $5x = 15$ $x = 3$ b) $3x = 15$ $x = 5$

Q4: a) $8 = 4x$ $x = 2$ b) $-9 = 3x$ $x = -3$ c) $6 = 2x$ $x = 3$

Q5: a) $3x = -12$ $x = -4$ b) $6 = 3x$ $x = 2$ c) $8 = 2x$ $x = 4$

Q6: a) $4x + 2 = 6x - 8$ b) $10 = 2x$ $x = 5$ c) $4(5) + 2 = 22$ sweets each

Q7: a) $10 + 2g = 4 + 4g$ b) $6 = 2g$ $g = 3$ GB c) cost = $10 + 2(3) = £16$

Q8: a) $5x - 3 = 2x + 9$ $3x = 12$ $x = 4$ b) Rod A = $5(4) - 3 = 17$ cm, Rod B = $2(4) + 9 = 17$ cm c) $2(17) - 5 = 29$ cm
